

Emergent Series Loading of a Plasma-Immersed Dipole by a Prescribed Vacuum Jacket in Three-Dimensional Fluid FDTD

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Key points

- A conductor-coated vacuum jacket in 3D fluid FDTD loads the feed as a series gap, not a shunt capacitor.
- In 1.50–1.80 MHz, thicker jackets make $\text{Im}\{Z\}$ more capacitive for all $S_d \geq 2$; near f_p that ordering reverses.
- A dipole coaxial estimate (two arms in series) matches band C_{eff} to within $\sim 20\%$; a fixed- Z_{pl} circuit overlay fails because $\text{Re}\{Z\}$ also drops with S_d .

Abstract

We quantify how a prescribed volumetric vacuum sheath of width S_d cells loads the feed impedance of a thin-wire dipole in a three-dimensional plasma-fluid finite-difference time-domain (PF-FDTD) solver. The sheath is a conductor-attached density hole: ambient density is depleted by 10^{-6} in a staircased shell of thickness $S_d\Delta x$ about every PEC cell, so plasma conduction current cannot reach the metal and feed current must close by displacement current across the radial gap. That geometry is the FDTD analogue of a series gap between metal and exterior plasma, not a shunt feed capacitance. Continuous-wave (CW) phasors at plasma frequency $f_p = 2$ MHz are reported for $S_d \in \{0, 2, 4, 6, 8, 10\}$ in 1.50–2.30 MHz, with a low-frequency extension at $S_d = 0, 2, 10$ (0.50–1.50 MHz). In 1.50–1.80 MHz, $\text{Im}\{Z\}$ at fixed frequency becomes steadily more negative as S_d increases for every $S_d \geq 2$. Near f_p that ordering reverses: thicker jackets damp the plasma-resonance swing rather than deepen capacitive reactance. Measured $\text{Im}\{Z\} + \rightarrow -$ crossings exist for $S_d = 0$ (1.846 MHz) and $S_d = 2$ (0.655 MHz, resting on a single inductive sample); $S_d = 4, 6, 8$ supply only upper bounds < 1.50 MHz, and $S_d = 10$ is capacitive throughout 0.50–2.30 MHz. Defining the mixed diagnostic $C_{\text{eff}} = -1/(\omega \text{Im}\{Z\})$ on capacitive branches and averaging 0.8–1.2 MHz gives $C_{\text{eff}}(2) \approx 7.4$ pF and $C_{\text{eff}}(10) \approx 4.3$ pF. A coaxial estimate for a *center-fed dipole* (two arms of length $L/2$ in series) yields $C_{\text{sh}}^{(\text{dip})} \approx 6.1$ and 3.7 pF for those widths, so $C_{\text{eff}}/C_{\text{sh}}^{(\text{dip})} \approx 1.2$. The Liu monopole formula overpredicts by a factor of four and is not the correct analytic object. A pedagogical circuit $Z(S_d = 0) + 1/(j\omega C_{\text{sh}}^{(\text{dip})})$ fails to reproduce the measured $\text{Re}\{Z\}(S_d)$ drop: the jacket both inserts a gap and removes near-wire plasma dielectric. The work is a static prescribed-jacket result on a small domain, not a claim of Tu/Song dynamic sheath fidelity.

Plain Language Summary

Radio antennas in space plasmas sit inside a thin empty sheath next to the metal. We simulated that empty layer as a jacket of controlled thickness around a dipole in a 3-D plasma wave code. Because the jacket wraps the wire, current has to jump the empty gap to reach the plasma; that is a series capacitor, not a capacitor across the antenna terminals. In a band below the plasma frequency, thicker jackets made the antenna look more capacitive for every thickness we ran. Near the plasma frequency the picture changes because the jacket also removes plasma near the wire. An analytic coax formula written for a dipole (not a monopole) matches the simulated capacitance to about twenty

percent in a low-frequency window, but the full impedance is not a fixed plasma load plus a pure capacitor.

1 Introduction

A conducting antenna immersed in plasma is surrounded by a depleted sheath that can dominate low-frequency impedance. Classical short-dipole theory in a cold magnetoplasma (Balmain, 1964, 1969) provides the homogeneous-medium baseline. Later treatments add ion sheaths, insulating jackets, and dielectric coatings as gap loading between metal and exterior plasma (Mlodnosky & Garriott, 1962; Galejs, 1966; Adachi et al., 1977). Modern plasma impedance probe (PIP) retrievals often write an explicit coaxial sheath capacitor in series with a Balmain-like plasma term (Brooks & Paliwoda, 2024; Liu et al., 2025), so that the phase-zero frequency lies below bulk plasma features.

High-voltage transmitting antennas raise a related but distinct problem: Song and coworkers modeled an oscillating ion-sheath radius about a bare dipole (Song et al., 2007), and Tu et al. used one-dimensional particle-in-cell (PIC) simulations to refine sheath structure and reactance (Tu et al., 2008). Those studies motivate the statement that sheath changes antenna impedance. They address self-consistent, time-varying sheaths on high-voltage transmitters; they do not prescribe a fixed vacuum jacket in three-dimensional fluid FDTD.

This paper asks a narrower question suited to the PF-FDTD solver used here (Ward, 2006): does a *prescribed* volumetric vacuum jacket of width $S_d\Delta x$ load a plasma-immersed dipole as an emergent *series* gap, and how does that loading compare with a dipole coaxial estimate? Success is defined by (i) monotonic capacitive ordering of $Z(f)$ with S_d in a stated band below f_p , (ii) honest reporting of resonance markers including upper bounds, and (iii) comparison to the correct dipole coax object after acknowledging that C_{eff} mixes sheath and plasma reactance. An upward Im-zero shift with S_d is not a success metric for this static-jacket experiment.

2 Methods

2.1 PF-FDTD solver and antenna geometry

PF-FDTD advances Maxwell’s equations on a staggered Yee grid coupled to multispecies fluid continuity and momentum equations (Ward, 2006). Plasma current $\mathbf{J} =$

$\sum_s q_s n_s \mathbf{u}_s$ enters Ampere's law. The present builds use $NS = 3$ species (electrons plus two positive-ion fluids with oxygen and nitric-oxide masses), cold fluid temperature $T = 0$, and a collision ratio $\nu/f_p = 0.1$ ($\nu = 200$ kHz at $f_p = 2$ MHz). Unless noted, the cyclotron frequency is set to zero (effectively unmagnetized). Bulk electron density follows from f_p : $N_{0,e} \approx 4.96 \times 10^{10} \text{ m}^{-3}$.

Feed observables are voltage $V(t)$ and current $I(t)$ at a hard E_z source. Voltage is $V = -E_z \Delta z$; current is the Ampere-loop circulation of $\mathbf{B}$ around the feed cell (Ward, 2006). Input impedance

$$Z(f) = \frac{V(f)}{I(f)} \quad (1)$$

is formed from steady-state CW phasors on the last 50% of each `.vc` record. Source type 3 (hard field) is imposed at the feed; this is not a soft gap feed and should be treated as part of the numerical antenna model.

All runs use a $70 \times 70 \times 65$ grid with $\Delta x = \Delta y = \Delta z = 0.04 \text{ m}$ (domain $\approx 2.8 \times 2.8 \times 2.6 \text{ m}$) and time step $\Delta t = 6.671114 \times 10^{-11} \text{ s}$. Domain faces use first-order retarded absorbing boundary conditions (Ward, 2006). A z -directed thin-wire dipole occupies grid indices (35, 35, 20) to (35, 35, 45) (length $L = 1.00 \text{ m}$) as a one-cell PEC line ($\text{ER}_z = 0$); the feed is at (35, 35, 33). No Holland–Simpson thin-wire radius correction is applied inside the solver; r_{eff} appears only in the analytic coax comparison below. Each CW tone is integrated for at least 10^5 steps and at least 15 drive cycles (`VcRate=10` samples). The domain is electrically small relative to $c/\omega_p \approx 24 \text{ m}$ and to free-space wavelengths at 0.5–2 MHz; absolute Z values may therefore carry box and boundary contributions, which is why trend-with- S_d is the primary claim.

2.2 Prescribed volumetric sheath as a series gap

The sheath is imposed as a *jacket that coats the conductor*, not as a circuit element connected across the feed terminals (Figure 1). Ambient density is initialized uniformly. After the antenna permittivity flags exist, a discrete distance field is grown from every PEC cell ($\text{ER}_x \text{ER}_y \text{ER}_z = 0$). Cells whose distance d (in grid steps) satisfies

$$0 < d \leq S_d \quad \Rightarrow \quad N_0 \rightarrow N_0 \cdot 10^{-6} \quad (2)$$

are depleted to a numerical floor ($N_{\text{min}}/N_0 = 10^{-6}$). Plasma conduction current in that shell is negligible, so the jacket behaves as vacuum. All results here use this PEC-seeded implementation. Figure 2 shows a *schematic* of the applied step, not a dumped field dump.

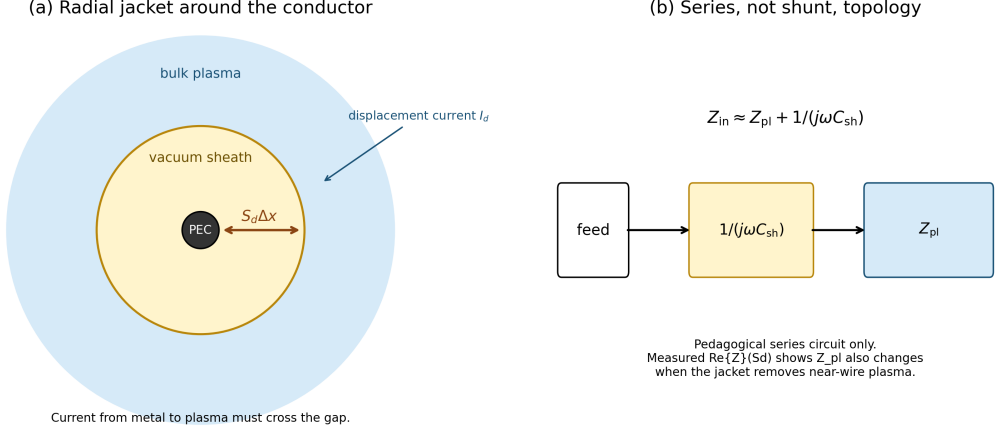


Figure 1. How the prescribed jacket mimics a series sheath. (a) Cross-section: a vacuum shell of thickness $S_d \Delta x$ coats the PEC wire; bulk plasma lies outside. (b) Equivalent topology: displacement current across the gap inserts $1/(j\omega C_{sh})$ in series with a plasma-loaded antenna path. A shunt capacitor would sit across the feed, which is not the geometry we impose. Measured $\text{Re}\{Z\}(S_d)$ shows that the plasma path is not invariant.

Because the depleted shell surrounds the metal, feed current reaching the bulk plasma must cross the radial gap as displacement current. A pedagogical circuit is

$$Z_{in}(f) \approx Z_{pl}(f) + \frac{1}{j\omega C_{sh}(S_d)}, \quad (3)$$

i.e. series sheath capacitance, not a shunt C across the terminals. Equation (3) is a useful topology statement, not an assumption that Z_{pl} is independent of S_d : thickening the jacket also removes plasma dielectric near the wire. Thin-wire geometry favors a logarithmic coaxial form for the gap estimate.

2.3 CW campaigns and resonance markers

Two CW grids share the same geometry and f_p :

Dense band (all thicknesses) $S_d \in \{0, 2, 4, 6, 8, 10\}$, $f = 1.50\text{--}2.30$ MHz (100 kHz and selected half steps; 66 completed cases). This is the full S_d survey and the primary loading evidence.

Low-frequency extension $S_d \in \{0, 2, 10\}$, $f = 0.50\text{--}1.50$ MHz (100 kHz steps). These three thicknesses were re-run below 1.50 MHz after the dense band showed that every $S_d \geq 2$ was already capacitive there. Intermediate widths $S_d = 4, 6, 8$ re-

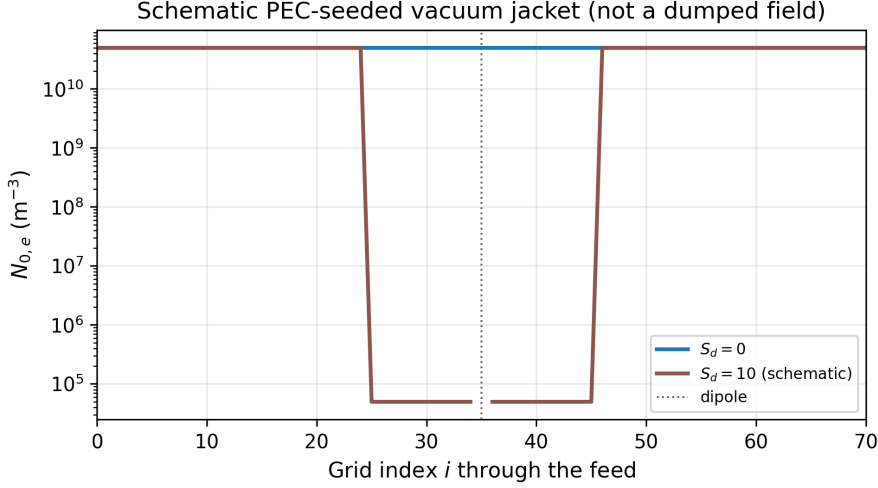


Figure 2. Schematic of the applied electron-density mask along a cut through the feed (not a dumped N_0 field). For $S_d = 10$ the PEC-seeded step depletes N_0 by 10^{-6} in ten cells on either side of the wire; the PEC cell itself carries no plasma density. $S_d = 0$ leaves a uniform plasma.

main in the dense-band results; their low-frequency crossings are not yet measured and are reported only as upper bounds.

One dense-band sample ($S_d = 0$, 1.50 MHz) is an incomplete restart file and is omitted from all plots and averages.

Series resonance f_{res} is the first $\text{Im}\{Z\}$ positive-to-negative zero crossing by linear interpolation when both neighboring samples exist. We also report $|Z|$ in the dense band as a second marker and do not equate phase-zero with magnitude extrema. On $\Delta x = 0.04$ m, $S_d = 2$ is already an 8 cm vacuum jacket—thick relative to a typical ionospheric Debye sheath at this density—so series-gap loading is intentionally strong.

2.4 Effective capacitance and coaxial references

Where $\text{Im}\{Z\} < 0$, define the mixed diagnostic

$$C_{\text{eff}}(f) = -\frac{1}{\omega \text{Im}\{Z(f)\}}, \quad \omega = 2\pi f. \quad (4)$$

This is an effective feed capacitance of the full antenna–sheath–plasma system on the present grid, not a pure vacuum gap to an ideal plasma electrode. It is frequency-dependent in the data and is used only as a secondary comparison, not as the primary claim.

The Liu-style coaxial sheath for a *monopole* of length L is

$$C_{\text{sh}}^{(\text{mono})} = \frac{2\pi\epsilon_0 L}{\ln(1 + S_d \Delta x / r_{\text{eff}})}. \quad (5)$$

For a center-fed dipole of total length L , each arm of length $L/2$ contributes half that capacitance if treated as an independent coax, and the two arms are electrically in series at the feed, so

$$C_{\text{sh}}^{(\text{dip})} = \frac{C_{\text{sh}}^{(\text{mono})}}{4} = \frac{\pi\epsilon_0 L}{2\ln(1 + S_d \Delta x / r_{\text{eff}})}. \quad (6)$$

We take $r_{\text{eff}} = 0.23\Delta x$ as a Holland-type thin-wire estimate for the analytic comparison only (Holland & Simpson, 1981), and also show $r_{\text{eff}} = 0.5\Delta x$ as a staircased-radius sensitivity. The quasi-static diagnostic average uses 0.8–1.2 MHz (capacitive points only) for $S_d = 2, 10$. Dense-band C_{eff} near f_p is not treated as a quasi-static capacitor.

3 Results

3.1 Dense-band impedance at all S_d

Figure 3 shows the full thickness survey. In 1.50–1.80 MHz, curves are ordered with S_d : larger jackets reduce $\text{Re}\{Z\}$ and drive $\text{Im}\{Z\}$ more negative at every frequency in that window. For $S_d \geq 2$, $\text{Im}\{Z\} < 0$ at all dense-band tones; only $S_d = 0$ crosses from inductive to capacitive, at 1.846 MHz. Near f_p (2.0–2.3 MHz) the ordering reverses: the unsheathed dipole develops the largest capacitive swing, and thicker jackets damp that plasma feature (Figure 4). Intermediate widths $S_d = 4, 6, 8$ lie between $S_d = 2$ and $S_d = 10$ in the low dense band; they are not outliers. The same low-band ordering appears in $|Z|$ (Figure 5).

3.2 Low-frequency extension for $S_d = 0, 2, 10$

Figure 6 stitches the low-frequency extension onto the dense band for the three thicknesses that were re-run below 1.50 MHz. $S_d = 0$ stays inductive through 0.50–1.50 MHz. $S_d = 10$ is capacitive at every low-frequency point. $S_d = 2$ is capacitive except for a brief inductive sample at 0.60 MHz, with an interpolated $+\rightarrow-$ crossing at 0.655 MHz. That crossing rests on a single inductive tone and should be treated as provisional pending denser sampling.

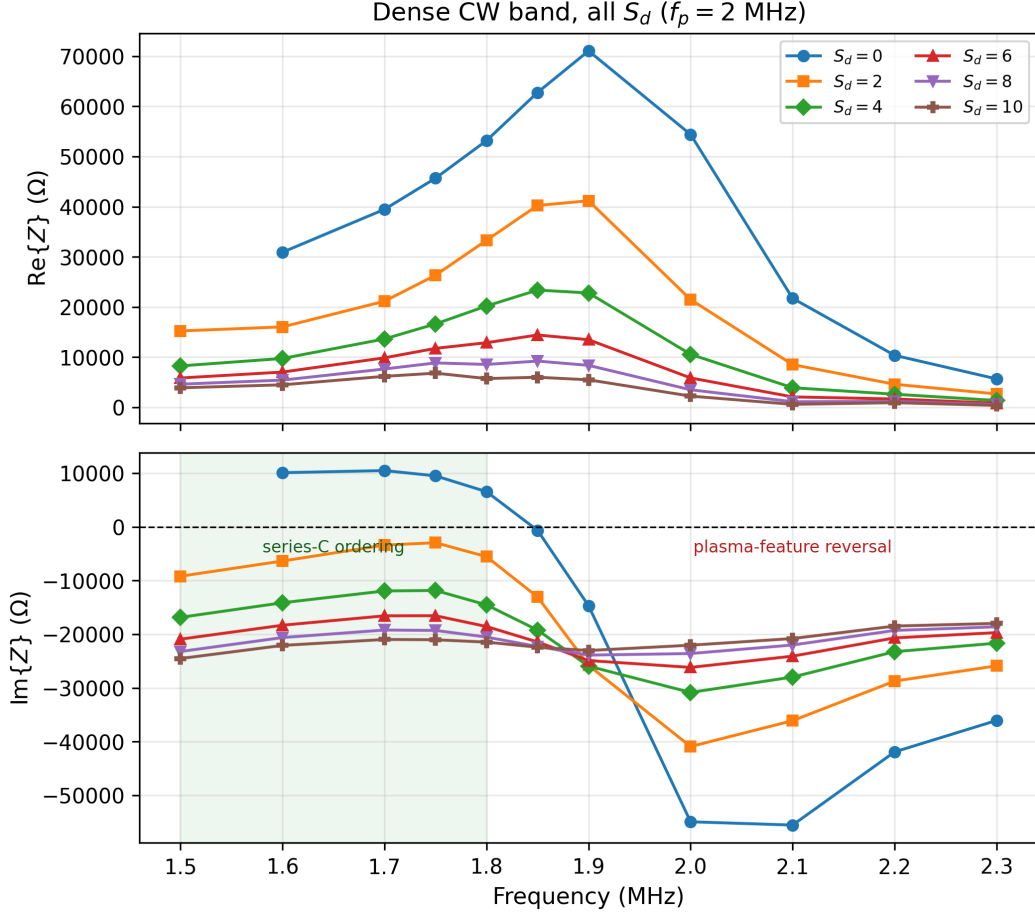


Figure 3. Dense-band CW phasor impedance for all simulated jackets $S_d = 0, 2, 4, 6, 8, 10$ ($f_p = 2$ MHz). Top: $\text{Re}\{Z\}$; bottom: $\text{Im}\{Z\}$. The $S_d = 0, 1.50$ MHz restart file is omitted. Green shading marks the band where thicker jackets are more capacitive; near f_p that ordering reverses.

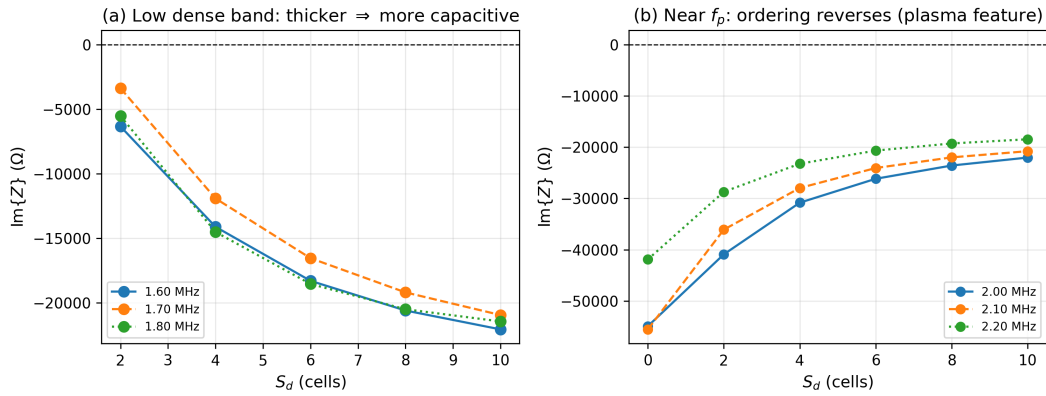


Figure 4. (a) Dense-band $\text{Im}\{Z\}(S_d)$ at 1.60, 1.70, and 1.80 MHz: monotonic capacitive loading with thickness. (b) Same quantity near f_p : thicker jackets reduce the plasma-resonance swing. This figure is the primary quantitative loading evidence.

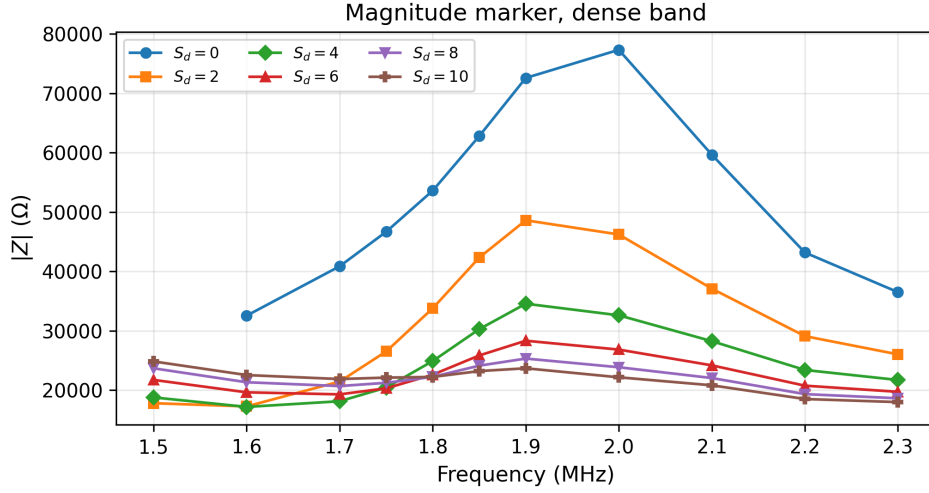


Figure 5. Dense-band $|Z|$ for the same six thicknesses. The near- f_p magnitude peak decreases monotonically with S_d and is a second marker, distinct from the $\text{Im}\{Z\} = 0$ crossing.

Table 1. Series resonance from $\text{Im}\{Z\} +$ to $-$ crossings

S_d	f_{res}	Note
0	1.846 MHz	Dense band; plasma-loaded resonance
2	0.655 MHz	Provisional; single inductive sample at 0.60 MHz
4, 6, 8	< 1.50 MHz	Upper bound only; no low-frequency run
10	≤ 0.50 MHz	All $\text{Im}\{Z\} < 0$ in 0.50–2.30 MHz

3.3 Resonance markers versus sheath width

Table 1 and Figure 7 collect $\text{Im}\{Z\} + \rightarrow -$ crossings. Measured crossings exist only for $S_d = 0$ and 2. For $S_d = 4, 6, 8$ the dense band supplies only an upper bound (< 1.50 MHz). $S_d = 10$ remains capacitive through 0.50 MHz. The plot therefore does not establish a continuous monotonic $f_{\text{res}}(S_d)$ curve across all six thicknesses; it establishes a downward move from $S_d = 0$ to $S_d = 2$ and bounds thereafter. The $S_d = 0$ marker is a plasma-loaded dipole resonance, not a pure sheath resonance, and should not be read as the same physical object as the sheathed crossings.

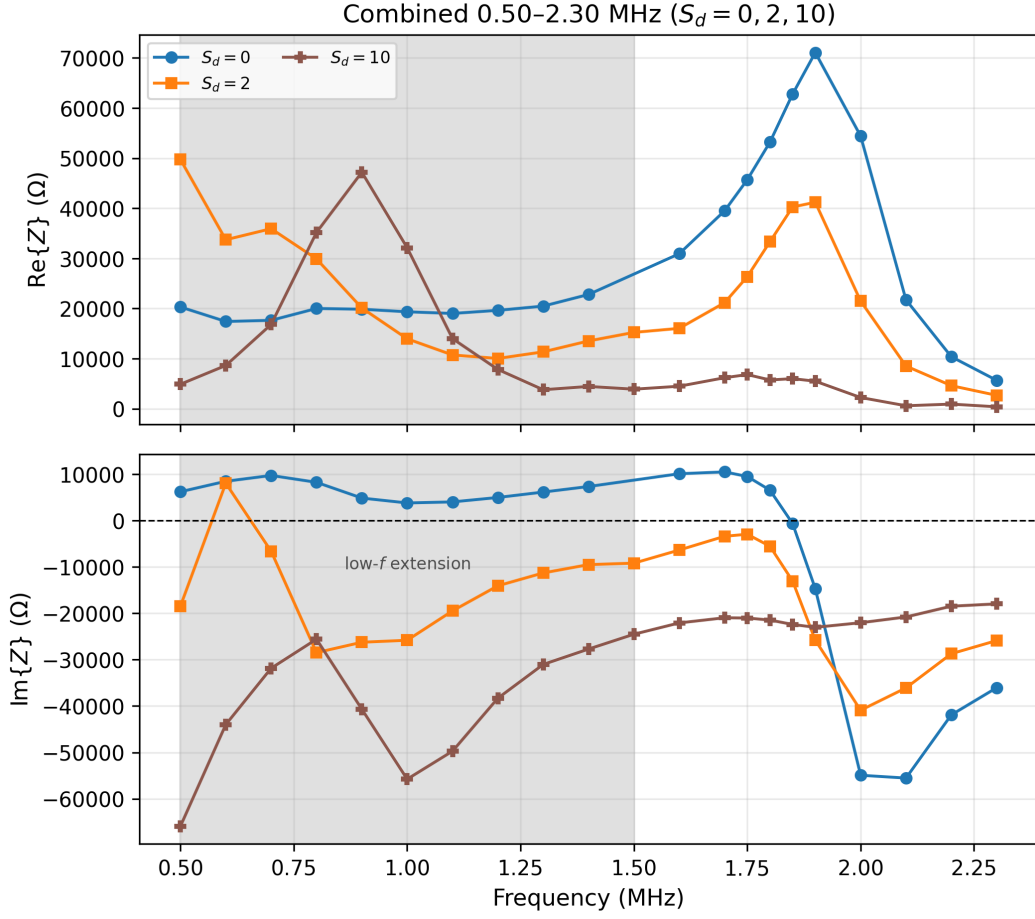


Figure 6. Combined CW impedance for $S_d = 0, 2, 10$ over 0.50–2.30 MHz. The shaded region is the low-frequency extension, run because the dense band (unshaded) already showed $S_d \geq 2$ to be capacitive.

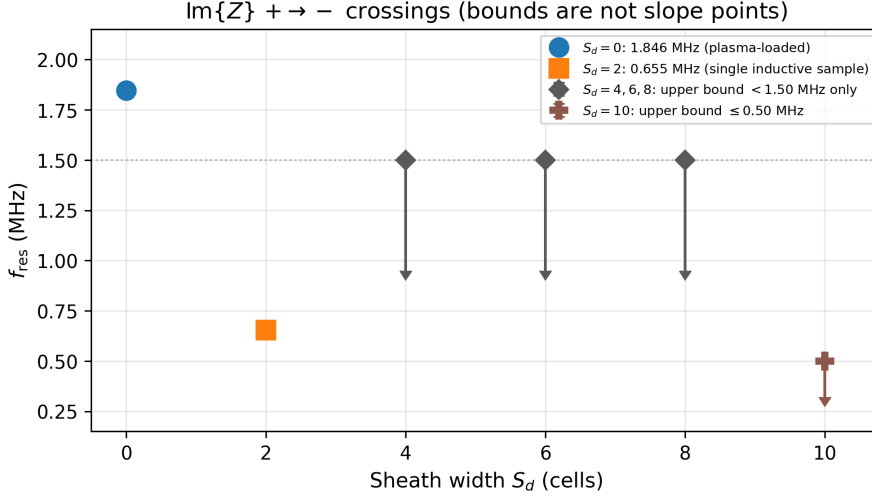


Figure 7. Resonance frequency $f_{\text{res}}(S_d)$ from $\text{Im}\{Z\}$ positive-to-negative crossings. Diamonds and the $S_d = 10$ symbol are upper limits (no crossing in the available band), not missing slope points.

Table 2. Diagnostic band-averaged C_{eff} (0.8–1.2 MHz) versus monopole and dipole coaxial references ($r_{\text{eff}} = 0.23\Delta x$).

S_d	t_{sh} (m)	C_{eff} (pF)	$C_{\text{sh}}^{(\text{mono})}$ (pF)	$C_{\text{sh}}^{(\text{dip})}$ (pF)	Ratio (mono)	Ratio (dip)
2	0.08	7.35 ± 1.12	24.49	6.12	0.30	1.20
10	0.40	4.27 ± 1.83	14.66	3.66	0.29	1.17

3.4 Capacitance diagnostic and circuit overlay

Figure 8 shows $C_{\text{eff}}(f)$ from equation (4). The quantity is not flat in the 0.8–1.2 MHz window: for $S_d = 10$ the band scatter is $\approx 43\%$ of the mean. Band means are therefore diagnostics, not high-precision capacitors (Table 2). Figure 9 compares those means with monopole and dipole coax formulae. After the dipole correction, $C_{\text{eff}}/C_{\text{sh}}^{(\text{dip})} \approx 1.20$ ($S_d = 2$) and 1.17 ($S_d = 10$). The monopole formula yields the previously reported factor ≈ 0.3 and overstates the analytic capacitance by four.

Figure 10 tests equation (3) with Z_{pl} fixed at the measured $S_d = 0$ impedance and $C_{\text{sh}} = C_{\text{sh}}^{(\text{dip})}$. Imaginary parts move in the right direction but do not match quantitatively. Real parts diverge strongly: measured $\text{Re}\{Z\}$ falls with S_d , whereas a lossless

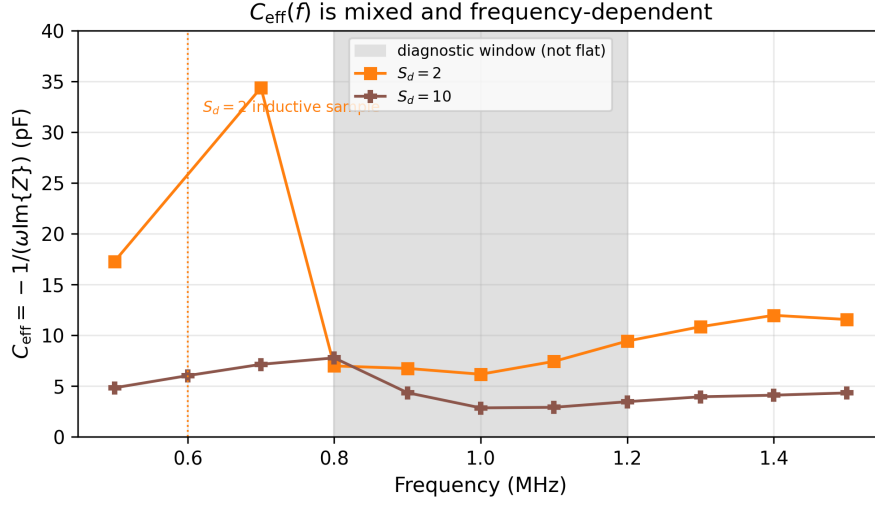


Figure 8. Effective capacitance $C_{\text{eff}}(f)$ for $S_d = 2$ and 10. The shaded band is the averaging window in Table 2. Frequency dependence shows that equation (4) is a mixed diagnostic, not a pure gap capacitor.

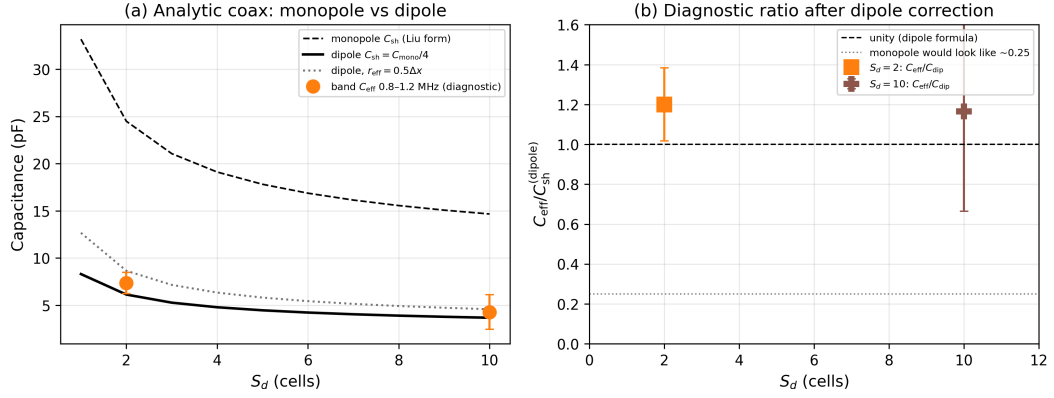


Figure 9. (a) Monopole coax (dashed), dipole coax (solid), and staircased-radius dipole sensitivity (dotted), with diagnostic low-frequency C_{eff} means. (b) Ratio to the dipole formula is order unity (≈ 1.2).

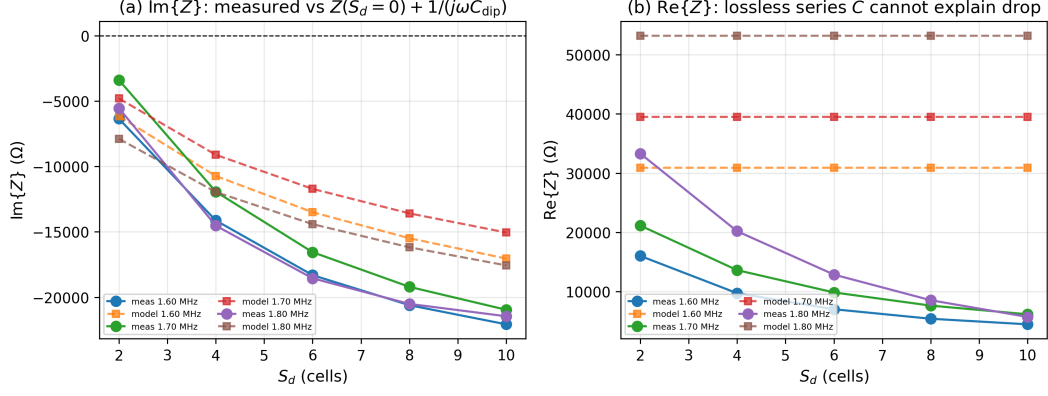


Figure 10. Circuit overlay at dense-band frequencies: measured $Z(S_d)$ versus $Z(S_d=0) + 1/(j\omega C_{\text{sh}}^{(\text{dip})})$. (a) Imaginary part. (b) Real part. Lossless series C cannot explain the measured resistive drop.

series capacitor leaves $\text{Re}\{Z\}$ unchanged. The jacket therefore cannot be reduced to a fixed plasma load plus a pure series C .

4 Discussion

4.1 Series versus parallel topology

The depleted shell sits between the conductor and bulk plasma (Figure 1). Series topology follows from that geometry. A shunt capacitor across the feed would not thicken when S_d increases around the wire, and it would not produce the ordered capacitive family in Figure 4a.

4.2 Why low-band capacitive loading strengthens with S_d

Thickening the jacket simultaneously (i) removes plasma dielectric near the wire and (ii) inserts series gap capacitance. In 1.50–1.80 MHz, the net effect is stronger capacitive $\text{Im}\{Z\}$. Near f_p , removing plasma dielectric and damping the resonance dominate, so ordering reverses (Figure 4b). On this grid $S_d = 2$ is already an 8 cm vacuum jacket—a strong load relative to a physical Debye sheath at ionospheric temperatures—which amplifies gap effects in the low band.

4.3 Analytic coax: monopole versus dipole

PIP formulations typically write equation (5) for a cylindrical monopole. Applying that formula unchanged to a center-fed dipole of total length L overpredicts feed capacitance by a factor of four. With equation (6), the diagnostic ratios become order unity. Remaining $\sim 20\%$ offsets are consistent with staircasing, r_{eff} ambiguity, residual plasma reactance inside C_{eff} , and the electrically small domain.

4.4 Relation to Song et al. and Tu et al.

Song et al. (2007) and Tu et al. (2008) address high-voltage transmitting dipoles with self-consistent, time-varying sheaths. The present runs prescribe a static step jacket in fluid FDTD. We therefore do not claim Tu dynamic fidelity. The appropriate static bridge is emergent gap loading and $C_{\text{sh}}(S_d)$, not an Im-zero scoreboard against those papers.

4.5 Limitations

1. Free-space f_{res} on the same dipole remains an open upper anchor (run plan documented with the analysis products).
2. Intermediate widths $S_d = 4, 6, 8$ lack low-frequency CW; f_{res} for those cases is only an upper bound. Denser tones around 0.55–0.70 MHz for $S_d = 2$ are also open.
3. C_{eff} mixes sheath and plasma reactance and is frequency-dependent; it is a diagnostic, not a de-embedded gap capacitance.
4. Absolute Z and C_{eff} may include box/ABC contributions on this electrically small domain.
5. Absolute coax comparison still depends on r_{eff} ; we report Holland-like $0.23\Delta x$ and staircased $0.5\Delta x$.
6. Campaigns here are effectively unmagnetized; PIP upper-hybrid markers require $B \neq 0$.
7. Cold fluid $T = 0$ has no thermal Debye length; physical-sheath comparisons use ionospheric T_e only as context.

5 Conclusions

A prescribed volumetric vacuum sheath in PF-FDTD loads a plasma-immersed dipole as a **series** gap because the jacket coats the wire and feed current must cross that gap by displacement current. In 1.50–1.80 MHz, CW measurements at all six thicknesses show monotonic capacitive loading with S_d . Near f_p that ordering reverses because the jacket also modifies the plasma path. Measured Im-zero crossings decrease from $S_d = 0$ to $S_d = 2$; intermediate widths are constrained only by upper bounds. A dipole coaxial estimate matches the low-frequency C_{eff} diagnostic to within $\sim 20\%$, whereas a monopole coax formula is the wrong object. A fixed- Z_{pl} series- C overlay fails for $\text{Re}\{Z\}$. These results establish emergent series-gap loading as the correct FDTD interpretation of a static jacket and provide a quantitative baseline—with stated limitations—before prescribing time-varying sheath boundaries or self-consistent charging.

Acknowledgments

The authors thank collaborators on the PF-FDTD sheath campaign for discussions of coupling diagnostics and impedance markers.

Open Research

Phasor tables and derived products supporting the figures are in `projects/tu2008-sheath/paper1_static`. Figure scripts are in the same analysis directory. Solver source is the collab PF-FDTD repository (PEC-seeded `ApplySheath`). Replace this section with a persistent DOI / repository link before submission.

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